MoleculeLens: Interpretable Drug–Text Retrieval via Salient Molecular Substructures

Anonymous Author(s)
Affiliation
Address
email

Abstract

Drug–text retrieval systems can connect molecular structures to mechanism descriptions, but most current models are difficult to audit. They often rely on large black-box encoders, and benchmark scores can be inflated by memorized scaffolds, drug names, or target metadata. As a result, it is unclear whether a retrieved match is supported by meaningful molecular substructures or by easier lexical shortcuts. We present *MoleculeLens*, an interpretable drug–text retrieval framework that makes each alignment decision traceable to salient molecular features. MoleculeLens aligns fixed molecular fingerprints and frozen biomedical text embeddings using learned linear projection heads. This simple design is the key contribution: the molecule-side linear map gives an efficient saliency score for each ECFP4 bit, allowing retrieval decisions to be inspected at the level of chemically meaningful substructures. On held-out scaffold pairs, this saliency proxy closely matches exact score-derived quantities, with mean Pearson correlation 0.905 against normalized-score gradients and 0.947 against leave-one-bit-out score drops. We evaluate MoleculeLens on a reproducible ChEMBL benchmark of 2,699 approved-drug mechanism pairs using a Bemis–Murcko scaffold-disjoint split. MoleculeLens achieves the strongest Recall@1 and MRR among the evaluated cross-modal baselines on this split, while also exposing when performance depends on drug-name or target/action metadata. Beyond retrieval accuracy, MoleculeLens provides pharmacophore-level explanations for correct matches, faithful diagnostics for failed retrievals, and a same-model substructure-level analysis of lexical leakage. These results show that lightweight linear bridges can make drug–text alignment more auditable without giving up competitive top-rank retrieval performance.

1 Introduction

Drug–text alignment asks whether molecular structures and mechanism descriptions can be embedded in a common space for retrieval. A useful system should generalize beyond memorized scaffolds and should reveal which molecular features support each match. Current evaluations make both questions difficult: drug names, target labels, and scaffold-correlated training examples can inflate retrieval scores, while standard dual encoders provide little evidence about the substructures behind a prediction.

MoleculeLens addresses these issues with a deliberately constrained architecture. It combines frozen ECFP4 fingerprints and a frozen biomedical text encoder with learned linear projection heads. The linear molecule-side bridge yields a closed-form saliency proxy over ECFP4 bits, which can be decoded with RDKit into atom-level substructures. We validate this proxy against exact gradients and exact leave-one-bit-out score drops, then use it to audit retrieval decisions and drug-name leakage under a scaffold-disjoint ChEMBL split.

Submitted to 40th Conference on Neural Information Processing Systems (NeurIPS 2026). Do not distribute.

This paper makes four contributions. First, we introduce validated closed-form linear saliency for drug–text retrieval. Second, we construct a reproducible scaffold-disjoint benchmark of 2,699 approved-drug mechanism pairs. Third, we show that leakage from explicit drug names is visible not only in retrieval scores but also in which molecular substructures the fixed rich model treats as salient. Fourth, we provide a descriptive contrastive logit-lens diagnostic showing that drug-retrievable CLS information in the frozen text encoder appears almost entirely at the final layer.

2 Related Work

Molecule–text retrieval and molecular representation learning. Drug–text and molecule–language models include retrieval-oriented systems such as Text2Mol [Edwards et al., 2021], molecular language models such as MolT5 [Edwards et al., 2022] and BioT5 [Pei et al., 2023], graph-language systems such as MolCA [Liu et al., 2023], and multimodal biomedical pretraining such as KV-PLM [Zeng et al., 2022]. Graphormer [Ying et al., 2021] and Uni-Mol [Zhou et al., 2023] are important molecular representation baselines, though not direct text-query retrieval systems without additional alignment. MoleculeLens is complementary: it emphasizes scaffold-disjoint drug–mechanism retrieval and auditable substructure-level explanations rather than maximum-capacity generation.

Leakage, scaffold splits, and interpretability. Scaffold splits are a standard guardrail against overoptimistic molecular generalization estimates [Bemis and Murcko, 1996]. We combine this with explicit drug-name and mechanism-only ablations because identity-bearing text fields can become shortcuts. The interpretability component is motivated by work on linear feature analysis and attribution graphs [Bricken et al., 2023, Templeton et al., 2024, Lindsey et al., 2025], but the features here are not discovered post hoc: ECFP4 bits already correspond to labelled circular molecular environments [Rogers and Hahn, 2010] recoverable with RDKit [RDKit contributors, 2023].

3 Benchmark and Method

Dataset and task. We construct the benchmark from ChEMBL [Zdrazil et al., 2024] by filtering for approved drugs (`max_phase=4`), valid mechanism text, and parseable canonical SMILES. The final dataset contains 2,699 drug–text pairs spanning 1,196 unique Bemis–Murcko scaffolds. The primary text field concatenates mechanism of action, target name, action type, and drug name:

$$\texttt{text_rich} = \texttt{mechanism} + \texttt{target} + \texttt{action} + \texttt{drug}. \tag{1}$$

The main split uses seed 0 over unique scaffolds: 2,019 train examples, 245 validation examples, and 435 held-out test examples whose scaffolds are unseen during training. We report text-query retrieval against a molecule gallery using single-label diagonal Recall@1, MRR, Recall@5, and Recall@10 for apples-to-apples comparison with baselines. Appendix A gives additional evaluation details, including the parent-structure handling used for attribution analyses.

Model. For drug i, RDKit computes a radius-2, 2048-bit ECFP4 fingerprint x_i [Rogers and Hahn, 2010]. For text i, a frozen biomedical sentence encoder produces a CLS-style embedding z_i. MoleculeLens learns only two linear projection heads:

$$b_i^{(m)} = W_m x_i + b_m, \qquad b_i^{(t)} = W_t z_i + b_t, \tag{2}$$

followed by L_2 normalization and cosine scoring. Training uses symmetric InfoNCE with same-target weighting and a hardest-same-target margin term. The scaffold model uses batch size 512, learning rate 10^{-3}, weight decay 10^{-4}, temperature 0.07, margin 0.1, and 100 epochs. All encoders are frozen; only W_m, b_m, W_t, b_t are trained. Architecture and implementation details are in Appendix B.

4 Retrieval Results

Table 1 shows the main comparison. MoleculeLens has the strongest scaffold-split Recall@1 and MRR among the evaluated cross-modal baselines, improving Recall@1 over MolPrompt by 9.6 points and over KV-PLM by 17.0 points. MolPrompt has higher Recall@5/10, so the claim is not

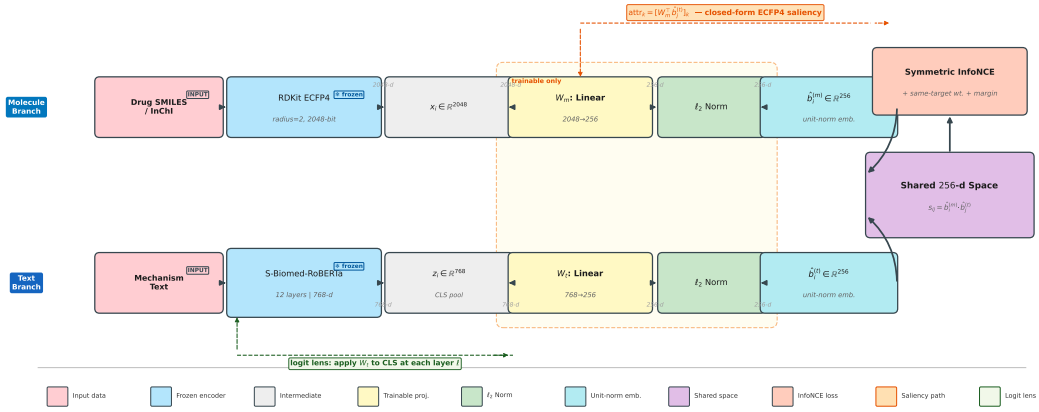

Figure 1: **MoleculeLens architecture.** Frozen ECFP4 and biomedical text encoders are connected by learned linear projection heads. The molecule-side linear map enables direct ECFP4-bit saliency; the text-side projection is reused for the layer-wise logit-lens diagnostic.

Table 1: Primary retrieval comparison. All rows use single-label diagonal evaluation. Full-gallery rows are context only; scaffold rows are the main comparison.

Model	Eval set	R@1	MRR	R@5	R@10
MoleculeSTM (zero-shot)	Full (N=2699)	0.089	0.171	0.243	0.337
Thin Bridges (Global)	Full (N=2699)	0.747	0.849	0.985	0.997
MolPrompt (Global)	Full (N=2699)	0.123	0.264	0.405	0.590
KV-PLM (Global)	Full (N=2699)	0.015	0.038	0.044	0.074
Graphormer (zero-shot)	Scaffold (N=435)	0.002	0.015	0.011	0.025
MoleculeLens	Scaffold (N=435)	**0.225**	**0.304**	0.384	0.462
MolPrompt	Scaffold (N=435)	0.129	0.256	**0.400**	**0.531**
KV-PLM	Scaffold (N=435)	0.055	0.121	0.172	0.251

leaderboard dominance: MoleculeLens is strongest at top-rank retrieval while remaining competitive at broader candidate coverage. Full-gallery rows are included only as context; the Thin Bridges global result is a memorization-heavy reference, not the main scientific comparison.

Text-field and leakage controls. Because `text_rich` includes target/action metadata and an explicit drug field, we report three complementary controls in Table 2. Removing only the drug field and retraining gives R@1 = 0.110, a 51.0% relative drop. Holding the rich model fixed and removing only `Drug: X.` at inference gives R@1 = 0.067, a stricter 70.4% same-model causal diagnostic. Training on mechanism text alone, removing target name, action type, and explicit drug name, gives R@1 = 0.124 and MRR = 0.226. Thus target/action fields contribute real signal, but mechanism-only retrieval remains nontrivial and close to MolPrompt scaffold R@1.

The non-overlapping Recall@1 intervals for MoleculeLens `text_rich` and MolPrompt support the reliability of the top-rank improvement despite the 435-example test set. We also reran the full MoleculeLens objective across the same five consecutive scaffold seeds used by the proxy robustness analysis. Table 3 shows that the complete objective is not weaker than the simplified proxy: mean R@1 is 0.277 ± 0.081 and mean MRR is 0.354 ± 0.083. The higher variance is expected because each seed induces a different scaffold-held-out test set, but every full-loss seed remains above KV-PLM scaffold R@1 and four of five exceed MolPrompt scaffold R@1. The three leakage numbers answer different questions: the 51.0% retrained drop is the fairest model-comparison number, the 70.4% same-model drop is the causal diagnostic for the fixed rich model, and the 22.9% multi-seed proxy mean measures partition sensitivity rather than canonical leakage.

Table 2: Key controls on the canonical scaffold split. Confidence intervals are 95% intervals for Recall@1; MoleculeLens uses exact bootstrap resampling and other baselines use Wilson intervals.

Condition	R@1	MRR	R@1 95% CI
MoleculeLens `text_rich`	0.225	0.304	[0.186, 0.267]
MoleculeLens retrained no-drug	0.110	–	[0.083, 0.140]
MoleculeLens same-model no-name	0.067	–	[0.044, 0.092]
MoleculeLens mechanism-only	0.124	0.226	–
MolPrompt scaffold	0.129	0.256	[0.101, 0.164]
KV-PLM scaffold	0.055	0.121	[0.037, 0.081]

Table 3: Five-seed full-loss check. These runs use the same same-target weighting and hardest-same-target margin term as the main MoleculeLens model. Seeds 0–4 were selected as the same consecutive seed set used by the proxy analysis before evaluating full-loss results.

Seed	n_{test}	R@1	R@5	R@10	MRR
0	403	0.221	0.335	0.424	0.291
1	261	0.295	0.479	0.533	0.381
2	224	0.259	0.375	0.442	0.328
3	221	0.407	0.561	0.629	0.484
4	395	0.203	0.342	0.451	0.284
Mean	–	**0.277**	**0.418**	**0.496**	**0.354**
Std	–	0.081	0.098	0.085	0.083

5 Substructure-Level Saliency

The implemented score is

$$s_{ij} = \hat{b}_i^{(m)} \cdot \hat{b}_j^{(t)} = \text{normalize}(W_m x_i + b_m) \cdot \text{normalize}(W_t z_j + b_t). \tag{3}$$

Because W_m is linear, we use the closed-form ECFP4 saliency proxy

$$\text{attr}_k(i, j) = \left[W_m^\top \hat{b}_j^{(t)}\right]_k, \tag{4}$$

where $\hat{b}_j^{(t)} = \text{normalize}(W_t z_j + b_t)$. Equation 4 is a fast ranking proxy rather than an exact causal decomposition: the exact normalized score also depends on the molecule-side bias and $\|W_m x_i + b_m\|$. We therefore validate it directly against exact gradients and exact leave-one-present-bit-out score drops over all 435 scaffold test pairs.

Table 4 supports rank-based use of the saliency proxy. It tracks the exact normalized-score gradient strongly and the exact bit-drop score even more closely. The identical-text test addresses a key conceptual concern: the exact top-10 bit-drop sets are almost never identical across distinct drugs sharing the same no-drug query, so the implemented normalized score is strongly drug-specific. The proxy also remains faithful for failed retrievals, which means incorrect predictions remain auditable decisions of the learned scoring function rather than cases where the attribution method collapses. Appendix D provides additional case studies.

What substructures recur? The saliency lens is useful only if it identifies interpretable molecular features rather than a single global shortcut. Table 5 summarizes the most frequent high-saliency ECFP4 bits across the scaffold test set. Bit 1538 is the most common, appearing in the top-10 for 85 pairs; RDKit decoding maps it to a benzene-ring environment with at least one heteroatom substituent at radius 2, a common pharmacophore anchor across several mechanism families. Frequencies drop sharply after the leading bits, and several frequent bits have near-zero mean signed attribution because they appear in both positive and negative contexts. This pattern supports the intended interpretation: MoleculeLens exposes recurring chemically meaningful substructures, but retrieval decisions are not dominated by one universal fingerprint bit.

Auditing near misses. Faithful attribution is useful beyond correctly retrieved examples. Among rank-2/3 errors after parent-structure filtering, 30 of 43 have attribution correlation above 0.5 between

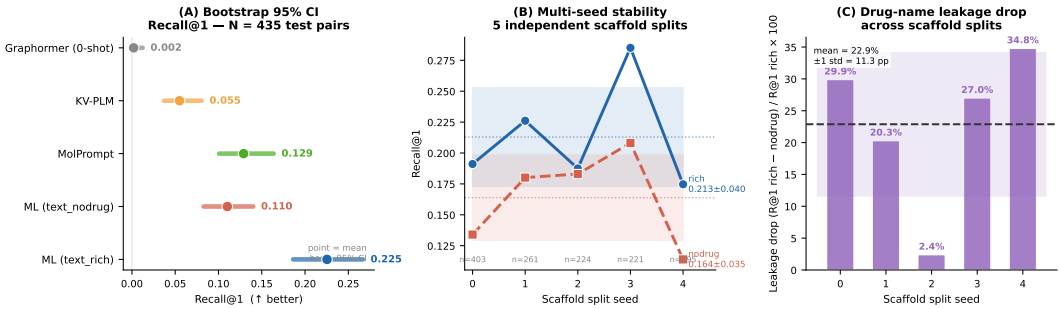

Figure 2: **Statistical robustness summary.** Bootstrap intervals, multi-seed scaffold variance, and per-seed drug-name leakage drops show that the main Recall@1 gap is statistically reliable while leakage varies with scaffold partition.

Table 4: Validation of Equation 4. All statistics are over present ECFP4 bits. The final rows test drug-specificity using distinct-drug pairs with identical no-drug text.

Metric	Value
Mean Pearson(attr, exact gradient)	0.905
Mean Pearson(attr, exact bit-drop)	0.947
Mean top-10 Jaccard(attr, exact bit-drop)	0.710
Incorrect Rank-1 pairs: Pearson(attr, bit-drop)	0.970
Incorrect Rank-1 pairs: top-10 Jaccard	0.761
Identical no-drug text: mean exact top-10 Jaccard	0.136
Identical no-drug text: non-identical exact top-10 sets	952/954

the correct drug and the retrieved drug, with mean correlation 0.569. These are not counted as correct in the diagonal retrieval metric, but the analysis distinguishes chemically plausible near misses from arbitrary failures. Figure 4 shows a representative case: sodium phenylacetate is ranked second while sodium phenylbutyrate is ranked first, and the two molecules share the same ammonia-lowering mechanism with highly overlapping salient substructures. This does not make the retrieval correct, but it shows that MoleculeLens can expose when an error is driven by related chemical evidence rather than an opaque embedding artifact.

Same-model leakage attribution. We next ask whether the same rich model uses the same substructures when the explicit drug name is removed at inference. For each test pair, let $\mathcal{T}_i^{\text{rich}}$ and $\mathcal{T}_i^{\text{nodrug}}$ be the top-10 saliency-bit sets under the original query and the same query with `Drug: X.` removed. We compute

$$J_i = \frac{|\mathcal{T}_i^{\text{rich}} \cap \mathcal{T}_i^{\text{nodrug}}|}{|\mathcal{T}_i^{\text{rich}} \cup \mathcal{T}_i^{\text{nodrug}}|}. \tag{5}$$

Mean overlap is only 0.356 ± 0.161; 82/435 pairs have $J_i < 0.2$, 36/435 have $J_i > 0.6$, and 13/435 are leakage candidates that are correct under `text_rich`, wrong under the same-model no-name query, and have $J_i < 0.2$. This converts the score-level leakage drop into a structural claim: for a substantial subset of test pairs, removing the name changes which molecular features drive retrieval even though model weights are fixed.

6 Contrastive Logit Lens

We adapt the logit-lens idea [nostalgebraist, 2020] by applying the trained text projection head W_t to the CLS state at each S-Biomed-RoBERTa layer. The diagnostic exactly replays the main rich model at layer 12: the reconstructed embeddings match the saved outputs within 1.21×10^{-3} maximum absolute error and recover R@1 = 0.225. Layers 0–11 are near chance for both rich and retrained no-drug text (R@1 between 0.002 and 0.011), while layer 12 jumps to R@1 = 0.225 for `text_rich` and 0.113 for `text_nodrug`. This is best viewed as a descriptive sanity check: retrieval signal and the rich-vs-no-name gap both appear in the final contextualized CLS representation, not gradually

Table 5: Most frequent high-saliency ECFP4 bits across 435 scaffold test pairs. Frequency is the number of test pairs for which the bit appears in the top-10 saliency ranking.

Bit index	Frequency	Mean attribution
1538	85 (19.5%)	0.228
1380	74 (17.0%)	0.004
1873	55 (12.6%)	0.000
807	44 (10.1%)	0.002
715	36 (8.3%)	0.029
350	35 (8.0%)	0.021
1171	33 (7.6%)	0.009
1917	32 (7.4%)	0.001
1039	31 (7.1%)	0.011
80	30 (6.9%)	0.000

across intermediate layers. The full layer table, family breakdown, and attention visualizations are in Appendix E.

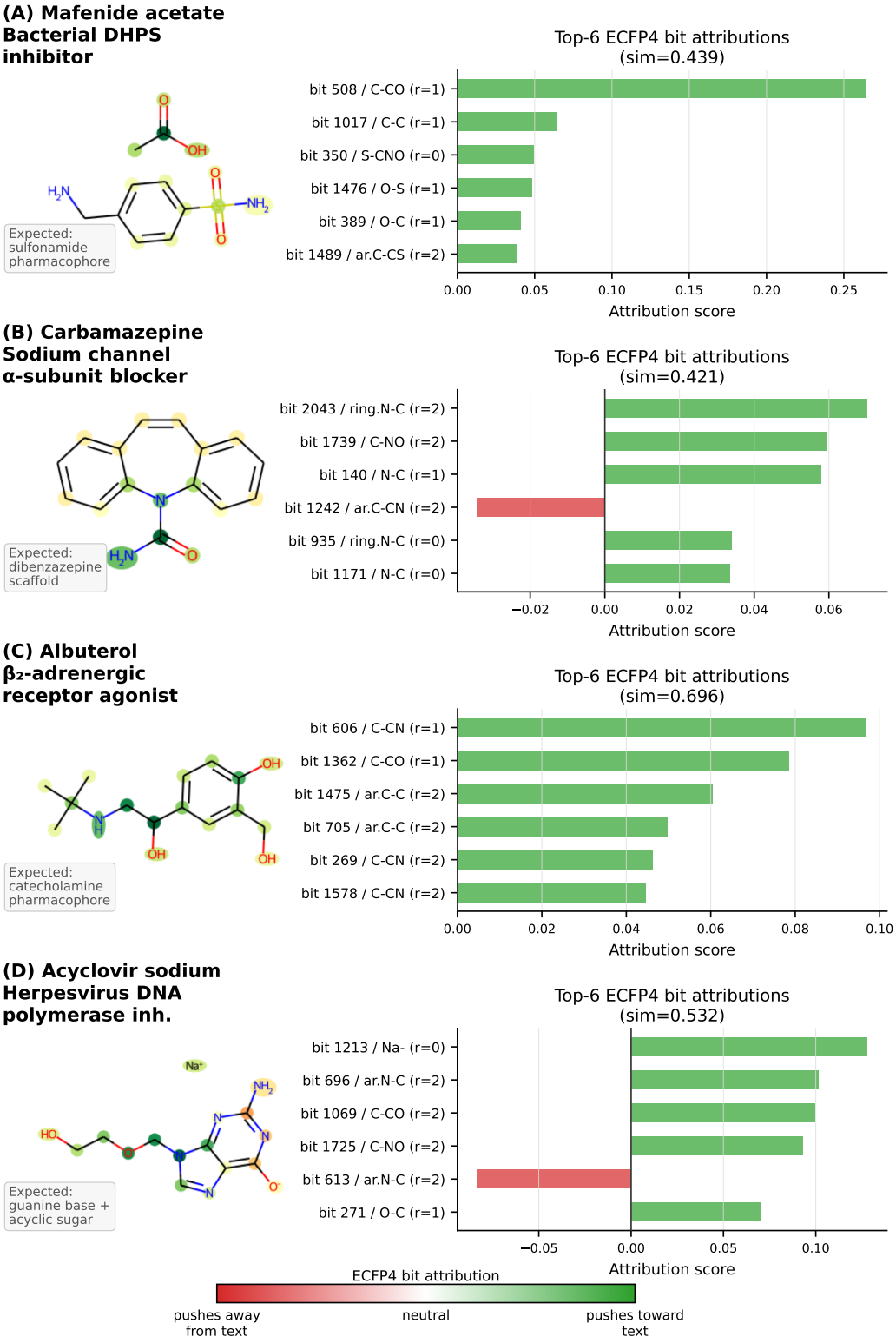

Figure 3: **ECFP4 linear saliency examples for correctly retrieved pairs.** Atoms are colour-coded by attribution value and paired with top-bit bar charts, showing that high-saliency bits often align with mechanism-relevant pharmacophoric regions.

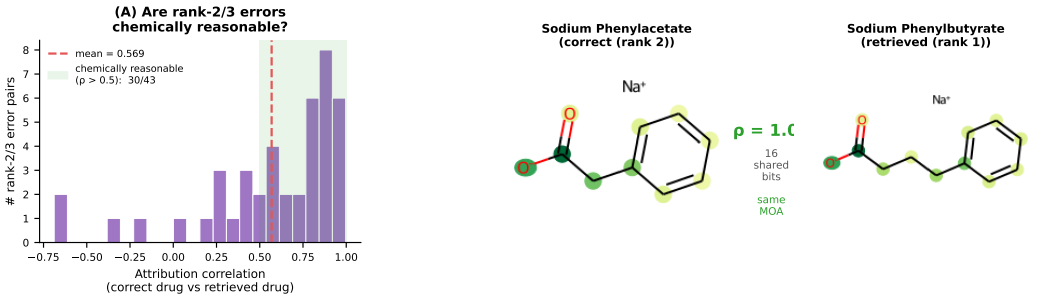

Figure 4: **Auditing wrong-but-close retrievals.** For rank-2/3 errors, attribution correlation separates chemically related near misses from less interpretable failures. Sodium phenylacetate and sodium phenylbutyrate illustrate a high-correlation same-mechanism confusion.

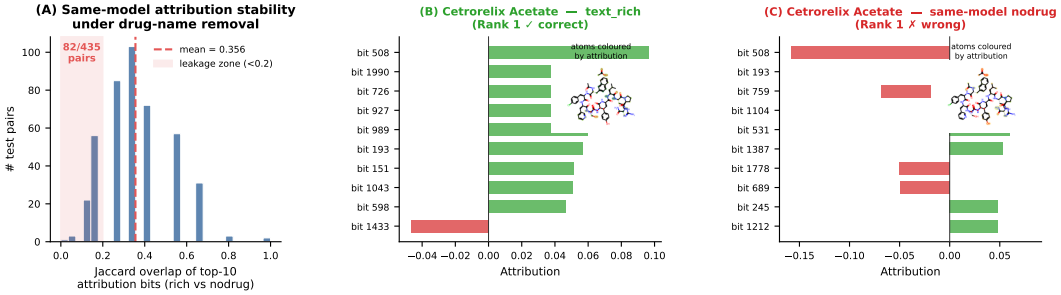

Figure 5: **Same-model drug-name ablation shifts salient substructures.** Removing only the explicit drug-name span from the query changes top-saliency ECFP4 bits for many test pairs; Cetrorelix acetate illustrates a low-overlap leakage candidate.

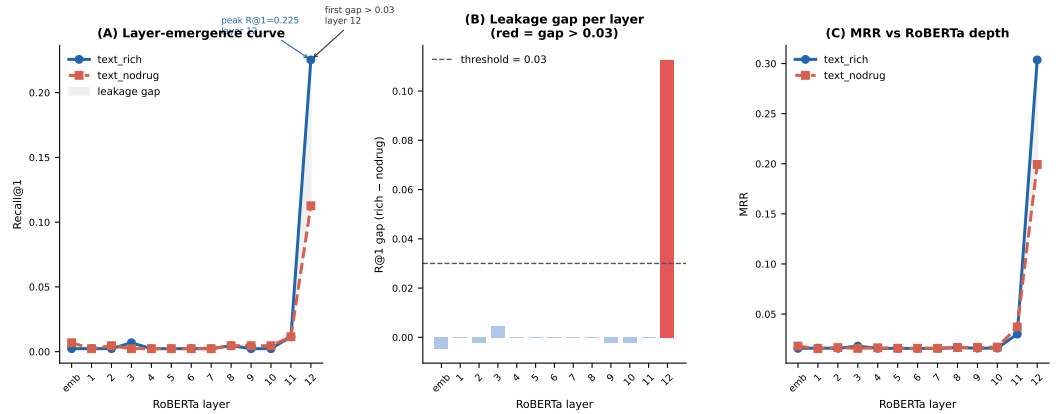

Figure 6: **Contrastive logit lens across RoBERTa layers.** Retrieval is near chance until the final transformer layer, where both rich retrieval signal and the no-name gap emerge.

7 Discussion and Limitations

MoleculeLens is not a claim that frozen ECFP4 plus a linear bridge solves scaffold-generalized drug–text retrieval. Its narrower contribution is that a lightweight model remains competitive at top-rank scaffold retrieval while making its decisions inspectable at the ECFP4-substructure level. The saliency proxy is validated against exact score-derived quantities, remains faithful on retrieval failures, and exposes same-model drug-name leakage as a shift in salient molecular features.

The benchmark has important limitations. The rich text field includes target and action metadata, and the mechanism-only ablation confirms that these fields contribute retrieval signal; the rich-text result should therefore be interpreted as curated mechanism-metadata retrieval rather than pure free-text mechanism understanding. The dataset covers FDA-approved drugs with curated ChEMBL mechanisms and does not establish generalization to investigational compounds or non-ChEMBL prose. Finally, the baseline set does not include adapted MolT5, BioT5, Uni-Mol, MolCA, or chemistry-LLM retrieval systems; these require nontrivial task adaptation and remain important future comparisons.

Reproducibility and intended use. The benchmark is designed to make these limitations visible rather than hide them. The scaffold split is released as a fixed artifact, the paper-facing tables are regenerated through a single artifact-refresh path, and all main numbers are tied to saved manifests. This matters because retrieval benchmarks are easy to contaminate: changing from scaffold-held-out evaluation to a full-gallery setting, or changing from same-model ablation to retrained no-drug comparison, answers a different scientific question. We therefore report the stronger rich-text result, the stricter mechanism-only condition, and the same-model no-name ablation side by side.

MoleculeLens is intended as an auditing tool for drug–text alignment, not as a standalone decision system for chemistry or medicine. It does not generate new molecules and should not be used for clinical or regulatory decisions without expert review. Its positive use case is narrower: when a retrieval model proposes a structure–mechanism match, MoleculeLens can show which fingerprint environments contributed to that match and whether those environments change under text perturbations. That makes it useful for diagnosing shortcut learning, prioritizing cases for medicinal-chemist inspection, and building stronger future benchmarks where higher-capacity encoders can be evaluated with the same scaffold and leakage controls.

8 Conclusion

On a 2,699-pair ChEMBL benchmark with scaffold-disjoint evaluation, MoleculeLens outperforms MolPrompt and KV-PLM on Recall@1 and MRR while clearly exposing the role of lexical and target/action metadata. Its central value is interpretability: the linear molecular bridge yields a validated, efficient ECFP4 saliency lens that can audit correct matches, failed retrievals, and same-model leakage. This makes MoleculeLens a useful foundation for future work on stronger encoders, text-identifiability audits, and feature-level analysis of drug–text alignment.

References

Guy W. Bemis and Mark A. Murcko. The properties of known drugs. 1. molecular frameworks. *Journal of Medicinal Chemistry*, 39(15):2887–2893, 1996.

Trenton Bricken, Adly Templeton, Joshua Batson, Brian Chen, Adam Jermyn, Tom Conerly, Nick Turner, Cem Anil, Carson Denison, Amanda Askell, Robert Lasenby, Yifan Wu, Shauna Kravec, Nicholas Schiefer, Tim Maxwell, Nicholas Joseph, Zac Hatfield-Dodds, Alex Tamkin, Karina Nguyen, Brayden McLean, Josh E Burke, Tristan Hume, Shan Carter, Tom Henighan, and Chris Olah. Towards monosemanticity: Decomposing language models with dictionary learning. `https://transformer-circuits.pub/2023/monosemantic-features/`, 2023.

Carl Edwards, Tuan Lai, Kevin Ros, Garrett Honke, and Heng Ji. Text2mol: Cross-modal molecule retrieval with natural language queries. In *Proceedings of the 2021 Conference on Empirical Methods in Natural Language Processing*, 2021.

Carl Edwards, Tuan Lai, Kevin Ros, Garrett Honke, Kyunghyun Cho, and Heng Ji. Translation between molecules and natural language. In *Proceedings of the 2022 Conference on Empirical Methods in Natural Language Processing*, pages 375–413, 2022. doi: 10.18653/v1/2022.emnlp-main.26.

Jack Lindsey, Wes Gurnee, Emmanuel Ameisen, Brian Chen, Adam Pearce, Nicholas L Turner, Craig Citro, Tom Conerly, Adly Templeton, Callum McDougall, et al. On the biology of a large language model. `https://transformer-circuits.pub/2025/attribution-graphs/methods.html`, 2025.

Zhiyuan Liu, Sihang Li, Yanchen Luo, Hao Fei, Yixin Cao, Kenji Kawaguchi, Xiang Wang, and Tat-Seng Chua. MolCA: Molecular graph-language modeling with cross-modal projector and uni-modal adapter. In *Proceedings of the 2023 Conference on Empirical Methods in Natural Language Processing*, pages 15623–15638, 2023. doi: 10.18653/v1/2023.emnlp-main.966.

nostalgebraist. Interpreting GPT: the logit lens. `https://www.lesswrong.com/posts/AcKRB8wDpdaN6v6ru/interpreting-gpt-the-logit-lens`, 2020.

Qizhi Pei, Wei Zhang, Jinhua Zhu, Kehan Wu, Kaiyuan Gao, Lijun Wu, Yingce Xia, and Rui Yan. BioT5: Enriching cross-modal integration in biology with chemical knowledge and natural language associations. In *Proceedings of the 2023 Conference on Empirical Methods in Natural Language Processing*, pages 1102–1123, 2023. doi: 10.18653/v1/2023.emnlp-main.70.

RDKit contributors. RDKit: Open-source cheminformatics. `https://www.rdkit.org`, 2023.

David Rogers and Mathew Hahn. Extended-connectivity fingerprints. *Journal of Chemical Information and Modeling*, 50(5):742–754, 2010. doi: 10.1021/ci100050t.

Adly Templeton, Tom Conerly, Jonathan Marcus, Jack Lindsey, Trenton Bricken, Brian Chen, Adam Pearce, Craig Citro, Emmanuel Ameisen, Andy Jones, Hoagy Cunningham, Nicholas L Turner, Callum McDougall, Monte MacDiarmid, C. Daniel Freeman, Theodore R. Sumers, Edward Rees, Joshua Batson, Adam Jermyn, Shan Carter, Chris Olah, and Tom Henighan. Scaling monosemanticity: Extracting interpretable features from Claude 3 Sonnet. `https://transformer-circuits.pub/2024/scaling-monosemanticity/`, 2024.

Chengxuan Ying et al. Do transformers really perform badly for graph representation? In *Advances in Neural Information Processing Systems*, 2021.

Barbara Zdrazil et al. The ChEMBL database in 2023: A drug discovery platform spanning multiple bioactivity data types and time periods. *Nucleic Acids Research*, 52(D1):D1180–D1192, 2024. doi: 10.1093/nar/gkad1004.

Zheni Zeng et al. A deep-learning system bridging molecule structure and biomedical text with comprehension comparable to human professionals. *Nature Communications*, 13(1):862, 2022. doi: 10.1038/s41467-022-28494-3.

Gengmo Zhou, Zhifeng Gao, Qiankun Ding, Hang Zheng, Hongteng Xu, Zhewei Wei, Linfeng Zhang, and Guolin Ke. Uni-Mol: A universal 3d molecular representation learning framework. In *International Conference on Learning Representations*, 2023.

A Additional Benchmark and Evaluation Details

The primary retrieval direction is text query to molecule gallery because this is the direction used by the attribution and logit-lens analyses. All main-table rows use single-label diagonal evaluation: a prediction is counted correct only when the exact matching row is returned. This is intentionally conservative for drugs with multiple ChEMBL mechanism rows, but it is the only consistent apples-to-apples protocol for baselines where per-example ranked lists are not available.

Several test drugs appear under multiple target–action rows. For attribution analyses, same-drug and wrong-but-close filtering use RDKit `FragmentParent` standardisation over SMILES to avoid treating salt or formulation variants as independent structures. In the canonical scaffold test set, 193 of 435 rows belong to 76 parent structures with multiple mechanism rows.

B Architecture and Implementation Details

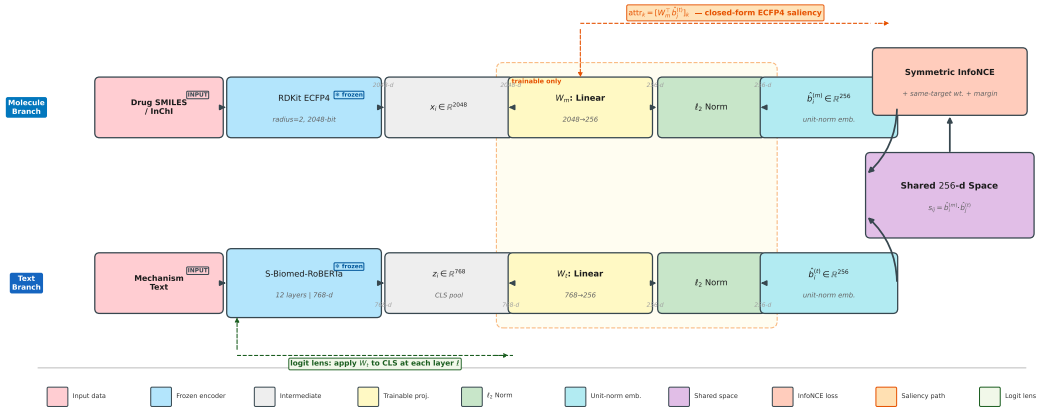

Figure 7: **MoleculeLens architecture.** A fixed ECFP4 molecule representation and a frozen biomedical text encoder are aligned by learned linear projection heads in a 256-dimensional shared space. Only the linear heads are trained; the encoders remain frozen. The molecule-side linear map enables the ECFP4 saliency proxy used in the main text, while the text-side projection head is reused for the contrastive logit-lens diagnostic.

The model trains for 100 epochs with batch size 512, learning rate 10^{-3}, weight decay 10^{-4}, temperature 0.07, and margin 0.1. Training the projection heads takes under 30 seconds per run on a single NVIDIA GPU. ECFP4 attribution and contrastive logit-lens analyses each complete in under 10 minutes.

C Robustness Details

Table 6: Multi-seed scaffold split variance for MoleculeLens using the simplified proxy objective. Five independent Bemis–Murcko splits are used; only linear projection heads are retrained per seed. Leakage drop is $(R@1_{\text{rich}} - R@1_{\text{nodrug}})/R@1_{\text{rich}}$.

Seed	n_{test}	R@1	R@5	R@10	MRR	Leakage drop
0	403	0.191	0.377	0.459	0.287	29.9%
1	261	0.226	0.479	0.544	0.342	20.3%
2	224	0.188	0.393	0.496	0.296	2.4%
3	221	0.285	0.525	0.634	0.402	27.0%
4	395	0.175	0.352	0.446	0.269	34.8%
Mean	–	**0.213**	**0.425**	**0.516**	**0.319**	**22.9%**
Std	–	0.045	0.070	0.073	0.053	12.6%

The corresponding full-loss five-seed check is reported in the main text in Table 3.

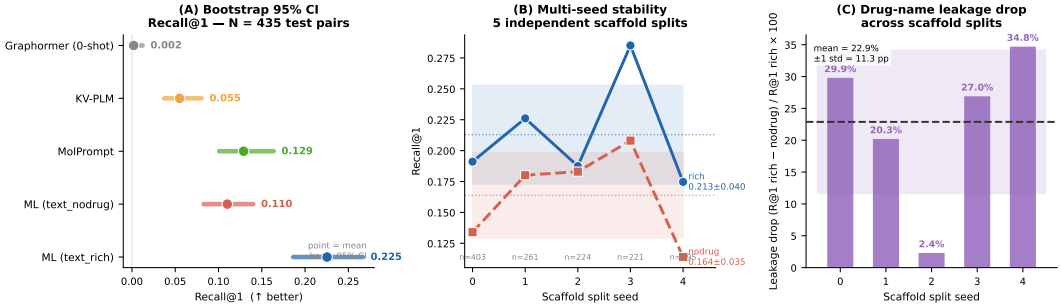

Figure 8: **Statistical robustness of MoleculeLens.** Bootstrap intervals, multi-seed scaffold variance, and per-seed drug-name leakage drops.

D Additional Attribution Analyses

For each test pair, we compute Equation 4, restrict to ECFP4 bits present in the molecule, rank bits by absolute saliency, and decode top-ranked bits with RDKit. Among eight correctly retrieved examples spanning antibacterial, antiviral, nuclear-receptor, opioid, sulfonamide, and diagnostic-agent mechanisms, the highest-saliency bits correspond to pharmacophoric or mechanism-defining regions such as steroid cores, sulfonamide groups, nucleoside analogue motifs, beta-lactam rings, and iodinated benzoate rings.

Aggregate high-saliency bit frequencies are reported in the main text in Table 5.

E Contrastive Logit-Lens Details

Table 7: Recall@1 and MRR at key RoBERTa layers under both text conditions. Layers 0–11 carry essentially no drug-retrievable signal; meaningful retrieval appears at layer 12.

Layer	text_rich		text_nodrug		Gap
	R@1	MRR	R@1	MRR	R@1
0	0.002	0.016	0.007	0.018	–0.005
3	0.007	0.018	0.002	0.016	0.005
6	0.002	0.016	0.002	0.016	0.000
9	0.002	0.016	0.005	0.017	–0.002
11	0.011	0.030	0.011	0.037	0.000
12	**0.225**	**0.304**	**0.113**	**0.199**	**0.113**

Table 8: Peak Recall@1 at layer 12 by mechanism family under `text_rich`.

Mechanism family	n	R@1
Nuclear Receptor agonist/antagonist	18	0.611
Kinase inhibitor	11	0.545
Antiviral (DNA polymerase inhibitor)	15	0.533
Antibacterial	25	0.440
Diagnostic agent	15	0.333
GPCR agonist/antagonist	129	0.271
COX Inhibitor	16	0.250
Ion Channel blocker	41	0.195
Other	165	0.152

F Limitations and Future Work

Dataset scope. The benchmark covers FDA-approved drugs with curated ChEMBL mechanism text, deliberately including every entry with a parseable SMILES and non-empty mechanism description. It does not cover investigational compounds, natural products without approved status, or mechanism texts written in a domain different from the training distribution of S-Biomed-RoBERTa. Generalisation to these settings is an open question.

Lexical leakage. Drug-name leakage is measured by removing the explicit `Drug: X.` field. Implicit lexical cues (e.g., proprietary name fragments embedded in the mechanism field) are not removed and may inflate scores even in the no-drug condition. The target name and action type fields in `text_rich` introduce an additional partial target-identity lookup signal beyond the explicit drug-name field; the mechanism-only ablation quantifies this effect but does not eliminate all possible lexical clustering.

Baseline scope. The primary comparison is restricted to cross-modal methods that score text queries against molecular representations. Same-modality text lookup controls are not included as competing baselines because they access gallery text rather than structure. Future benchmark releases should include a separate text-identifiability audit, but those diagnostics should be reported separately from cross-modal retrieval baselines.

SAE and cross-modal crosscoder. The shared 256-dimensional embedding space is amenable to sparse autoencoder analysis [Templeton et al., 2024, Bricken et al., 2023] to measure whether alignment dimensions are monosemantic or polysemantic. A cross-modal crosscoder [Lindsey et al., 2025] trained on concatenated drug–text embedding pairs would further distinguish cross-modal alignment features from text-only leakage features. These analyses are left for future work.

G Broader Impacts

Broader impacts and intended-use considerations are discussed in the main text. In brief, MoleculeLens is an auditing framework for publicly available ChEMBL drug–mechanism retrieval and is not intended for clinical or regulatory deployment without expert oversight.

NeurIPS Paper Checklist

1. **Claims**

   Question: Do the main claims made in the abstract and introduction accurately reflect the paper's contributions and scope?

   Answer: [Yes]

   Justification: The abstract and introduction state four specific contributions (validated closed-form linear saliency framework, leakage anatomy, descriptive contrastive logit-lens diagnostic, reproducible scaffold benchmark) that are each demonstrated experimentally in the corresponding sections. The scope is explicitly limited to FDA-approved drugs with ChEMBL mechanism text under scaffold-disjoint evaluation.

   Guidelines:

   - The answer [N/A] means that the abstract and introduction do not include the claims made in the paper.
   - The abstract and/or introduction should clearly state the claims made, including the contributions made in the paper and important assumptions and limitations. A [No] or [N/A] answer to this question will not be perceived well by the reviewers.
   - The claims made should match theoretical and experimental results, and reflect how much the results can be expected to generalize to other settings.
   - It is fine to include aspirational goals as motivation as long as it is clear that these goals are not attained by the paper.

2. **Limitations**

Question: Does the paper discuss the limitations of the work performed by the authors?

Answer: [Yes]

Justification: The Discussion (Section 9) and the Limitations and Future Work appendix cover: (i) dataset scope limited to FDA-approved drugs; (ii) residual implicit lexical leakage not captured by the drug-name ablation; (iii) target/action metadata leakage quantified by the mechanism-only ablation; (iv) the restriction of the primary comparison to cross-modal structure-aware baselines; (v) missing adapted evaluations of newer molecular language models; (vi) domain-specificity and the open question of generalization to unapproved compounds.

Guidelines:

- The answer [N/A] means that the paper has no limitation while the answer [No] means that the paper has limitations, but those are not discussed in the paper.
- The authors are encouraged to create a separate "Limitations" section in their paper.
- The paper should point out any strong assumptions and how robust the results are to violations of these assumptions (e.g., independence assumptions, noiseless settings, model well-specification, asymptotic approximations only holding locally). The authors should reflect on how these assumptions might be violated in practice and what the implications would be.
- The authors should reflect on the scope of the claims made, e.g., if the approach was only tested on a few datasets or with a few runs. In general, empirical results often depend on implicit assumptions, which should be articulated.
- The authors should reflect on the factors that influence the performance of the approach. For example, a facial recognition algorithm may perform poorly when image resolution is low or images are taken in low lighting. Or a speech-to-text system might not be used reliably to provide closed captions for online lectures because it fails to handle technical jargon.
- The authors should discuss the computational efficiency of the proposed algorithms and how they scale with dataset size.
- If applicable, the authors should discuss possible limitations of their approach to address problems of privacy and fairness.
- While the authors might fear that complete honesty about limitations might be used by reviewers as grounds for rejection, a worse outcome might be that reviewers discover limitations that aren't acknowledged in the paper. The authors should use their best judgment and recognize that individual actions in favor of transparency play an important role in developing norms that preserve the integrity of the community. Reviewers will be specifically instructed to not penalize honesty concerning limitations.

3. **Theory assumptions and proofs**

Question: For each theoretical result, does the paper provide the full set of assumptions and a complete (and correct) proof?

Answer: [N/A]

Justification: The paper contains no formal theorems. Equation 2 is presented as a closed-form linear saliency approximation derived directly from the linear projection heads; its approximation status and omitted bias/normalization terms are stated explicitly in Section 7.1, so no separate proof is required.

Guidelines:

- The answer [N/A] means that the paper does not include theoretical results.
- All the theorems, formulas, and proofs in the paper should be numbered and cross-referenced.
- All assumptions should be clearly stated or referenced in the statement of any theorems.
- The proofs can either appear in the main paper or the supplemental material, but if they appear in the supplemental material, the authors are encouraged to provide a short proof sketch to provide intuition.
- Inversely, any informal proof provided in the core of the paper should be complemented by formal proofs provided in appendix or supplemental material.

- Theorems and Lemmas that the proof relies upon should be properly referenced.

4. **Experimental result reproducibility**

Question: Does the paper fully disclose all the information needed to reproduce the main experimental results of the paper to the extent that it affects the main claims and/or conclusions of the paper (regardless of whether the code and data are provided or not)?

Answer: [Yes]

Justification: Section 3 describes the ChEMBL data construction (max_phase filter, SMILES fetching, Bemis–Murcko scaffold split), Section 5.2 gives all hyperparameters (batch size, LR, weight decay, temperature, margin, epochs), and Section 7.1 states the linear saliency formula and its approximation assumptions. The repo now includes a single refresh command, `bash scripts/run_paper_artifact_refresh.sh`, which rewrites the comparison manifest, canonical Track 1/Track 3 artifacts, paper figures, and a machine-readable `results/paper_artifact_manifest.json` that records the source files and key numbers backing the manuscript.

Guidelines:

- The answer [N/A] means that the paper does not include experiments.
- If the paper includes experiments, a [No] answer to this question will not be perceived well by the reviewers: Making the paper reproducible is important, regardless of whether the code and data are provided or not.
- If the contribution is a dataset and/or model, the authors should describe the steps taken to make their results reproducible or verifiable.
- Depending on the contribution, reproducibility can be accomplished in various ways. For example, if the contribution is a novel architecture, describing the architecture fully might suffice, or if the contribution is a specific model and empirical evaluation, it may be necessary to either make it possible for others to replicate the model with the same dataset, or provide access to the model. In general. releasing code and data is often one good way to accomplish this, but reproducibility can also be provided via detailed instructions for how to replicate the results, access to a hosted model (e.g., in the case of a large language model), releasing of a model checkpoint, or other means that are appropriate to the research performed.
- While NeurIPS does not require releasing code, the conference does require all submissions to provide some reasonable avenue for reproducibility, which may depend on the nature of the contribution. For example
  (a) If the contribution is primarily a new algorithm, the paper should make it clear how to reproduce that algorithm.
  (b) If the contribution is primarily a new model architecture, the paper should describe the architecture clearly and fully.
  (c) If the contribution is a new model (e.g., a large language model), then there should either be a way to access this model for reproducing the results or a way to reproduce the model (e.g., with an open-source dataset or instructions for how to construct the dataset).
  (d) We recognize that reproducibility may be tricky in some cases, in which case authors are welcome to describe the particular way they provide for reproducibility. In the case of closed-source models, it may be that access to the model is limited in some way (e.g., to registered users), but it should be possible for other researchers to have some path to reproducing or verifying the results.

5. **Open access to data and code**

Question: Does the paper provide open access to the data and code, with sufficient instructions to faithfully reproduce the main experimental results, as described in supplemental material?

Answer: [No]

Justification: Code and benchmark data (scaffold split manifest, projection weights, repo-root README, and the canonical `scripts/run_paper_artifact_refresh.sh` entry-point) will be released publicly upon acceptance of the paper. The data is derived entirely from ChEMBL, which is publicly available under CC BY-SA 3.0.

Guidelines:

- The answer [N/A] means that paper does not include experiments requiring code.
- Please see the NeurIPS code and data submission guidelines (`https://neurips.cc/public/guides/CodeSubmissionPolicy`) for more details.
- While we encourage the release of code and data, we understand that this might not be possible, so [No] is an acceptable answer. Papers cannot be rejected simply for not including code, unless this is central to the contribution (e.g., for a new open-source benchmark).
- The instructions should contain the exact command and environment needed to run to reproduce the results. See the NeurIPS code and data submission guidelines (`https://neurips.cc/public/guides/CodeSubmissionPolicy`) for more details.
- The authors should provide instructions on data access and preparation, including how to access the raw data, preprocessed data, intermediate data, and generated data, etc.
- The authors should provide scripts to reproduce all experimental results for the new proposed method and baselines. If only a subset of experiments are reproducible, they should state which ones are omitted from the script and why.
- At submission time, to preserve anonymity, the authors should release anonymized versions (if applicable).
- Providing as much information as possible in supplemental material (appended to the paper) is recommended, but including URLs to data and code is permitted.

6. **Experimental setting/details**

Question: Does the paper specify all the training and test details (e.g., data splits, hyperparameters, how they were chosen, type of optimizer) necessary to understand the results?

Answer: [Yes]

Justification: Section 3.3 specifies the Bemis–Murcko scaffold split sizes (2,019 train / 245 val / 435 test), and Section 5.2 lists all training hyperparameters. The Adam optimizer is used by default; the text encoder is frozen throughout.

Guidelines:

- The answer [N/A] means that the paper does not include experiments.
- The experimental setting should be presented in the core of the paper to a level of detail that is necessary to appreciate the results and make sense of them.
- The full details can be provided either with the code, in appendix, or as supplemental material.

7. **Experiment statistical significance**

Question: Does the paper report error bars suitably and correctly defined or other appropriate information about the statistical significance of the experiments?

Answer: [Yes]

Justification: Table 2 reports bootstrap 95% confidence intervals (10,000 resamples) for Recall@1 for MoleculeLens and Wilson score intervals for baselines. Table 4 reports mean and standard deviation of Recall@1 across five independent scaffold splits. The factors of variability (test-set resampling; scaffold partition randomness) are described in Sections 6.1 and 6.1 respectively.

Guidelines:

- The answer [N/A] means that the paper does not include experiments.
- The authors should answer [Yes] if the results are accompanied by error bars, confidence intervals, or statistical significance tests, at least for the experiments that support the main claims of the paper.
- The factors of variability that the error bars are capturing should be clearly stated (for example, train/test split, initialization, random drawing of some parameter, or overall run with given experimental conditions).
- The method for calculating the error bars should be explained (closed form formula, call to a library function, bootstrap, etc.)

- The assumptions made should be given (e.g., Normally distributed errors).
- It should be clear whether the error bar is the standard deviation or the standard error of the mean.
- It is OK to report 1-sigma error bars, but one should state it. The authors should preferably report a 2-sigma error bar than state that they have a 96% CI, if the hypothesis of Normality of errors is not verified.
- For asymmetric distributions, the authors should be careful not to show in tables or figures symmetric error bars that would yield results that are out of range (e.g., negative error rates).
- If error bars are reported in tables or plots, the authors should explain in the text how they were calculated and reference the corresponding figures or tables in the text.

8. **Experiments compute resources**

   Question: For each experiment, does the paper provide sufficient information on the computer resources (type of compute workers, memory, time of execution) needed to reproduce the experiments?

   Answer: [Yes]

   Justification: Section 5.2 states that all experiments run on a single NVIDIA GPU, training completes in under 30 seconds per run, and the attribution and logit-lens analyses each complete in under 10 minutes. The dominant compute cost is the one-time RoBERTa text encoding, which is shared across all experiments.

   Guidelines:

   - The answer [N/A] means that the paper does not include experiments.
   - The paper should indicate the type of compute workers CPU or GPU, internal cluster, or cloud provider, including relevant memory and storage.
   - The paper should provide the amount of compute required for each of the individual experimental runs as well as estimate the total compute.
   - The paper should disclose whether the full research project required more compute than the experiments reported in the paper (e.g., preliminary or failed experiments that didn't make it into the paper).

9. **Code of ethics**

   Question: Does the research conducted in the paper conform, in every respect, with the NeurIPS Code of Ethics `https://neurips.cc/public/EthicsGuidelines`?

   Answer: [Yes]

   Justification: The research uses publicly available ChEMBL data (CC BY-SA 3.0), does not involve human subjects, does not generate potentially harmful content, and raises no privacy or fairness concerns.

   Guidelines:

   - The answer [N/A] means that the authors have not reviewed the NeurIPS Code of Ethics.
   - If the authors answer [No], they should explain the special circumstances that require a deviation from the Code of Ethics.
   - The authors should make sure to preserve anonymity (e.g., if there is a special consideration due to laws or regulations in their jurisdiction).

10. **Broader impacts**

    Question: Does the paper discuss both potential positive societal impacts and negative societal impacts of the work performed?

    Answer: [Yes]

    Justification: The Broader Impacts appendix discusses positive impacts (transparent, auditable drug-discovery retrieval tools; surfacing evaluation pitfalls) and negative impacts (the framework is not intended for clinical deployment and raises no significant negative risk).

    Guidelines:

- The answer [N/A] means that there is no societal impact of the work performed.
- If the authors answer [N/A] or [No], they should explain why their work has no societal impact or why the paper does not address societal impact.
- Examples of negative societal impacts include potential malicious or unintended uses (e.g., disinformation, generating fake profiles, surveillance), fairness considerations (e.g., deployment of technologies that could make decisions that unfairly impact specific groups), privacy considerations, and security considerations.
- The conference expects that many papers will be foundational research and not tied to particular applications, let alone deployments. However, if there is a direct path to any negative applications, the authors should point it out. For example, it is legitimate to point out that an improvement in the quality of generative models could be used to generate Deepfakes for disinformation. On the other hand, it is not needed to point out that a generic algorithm for optimizing neural networks could enable people to train models that generate Deepfakes faster.
- The authors should consider possible harms that could arise when the technology is being used as intended and functioning correctly, harms that could arise when the technology is being used as intended but gives incorrect results, and harms following from (intentional or unintentional) misuse of the technology.
- If there are negative societal impacts, the authors could also discuss possible mitigation strategies (e.g., gated release of models, providing defenses in addition to attacks, mechanisms for monitoring misuse, mechanisms to monitor how a system learns from feedback over time, improving the efficiency and accessibility of ML).

11. **Safeguards**

Question: Does the paper describe safeguards that have been put in place for responsible release of data or models that have a high risk for misuse (e.g., pre-trained language models, image generators, or scraped datasets)?

Answer: [N/A]

Justification: The model (two small linear projection matrices) and the benchmark (a filtered subset of public ChEMBL data) pose no meaningful misuse risk.

Guidelines:

- The answer [N/A] means that the paper poses no such risks.
- Released models that have a high risk for misuse or dual-use should be released with necessary safeguards to allow for controlled use of the model, for example by requiring that users adhere to usage guidelines or restrictions to access the model or implementing safety filters.
- Datasets that have been scraped from the Internet could pose safety risks. The authors should describe how they avoided releasing unsafe images.
- We recognize that providing effective safeguards is challenging, and many papers do not require this, but we encourage authors to take this into account and make a best faith effort.

12. **Licenses for existing assets**

Question: Are the creators or original owners of assets (e.g., code, data, models), used in the paper, properly credited and are the license and terms of use explicitly mentioned and properly respected?

Answer: [Yes]

Justification: ChEMBL [Zdrazil et al., 2024] is used under CC BY-SA 3.0. RDKit [RDKit contributors, 2023] is used under the BSD license. S-Biomed-RoBERTa is used for inference only. KV-PLM [Zeng et al., 2022], MolPrompt, and Graphormer [Ying et al., 2021] are cited and evaluated under their respective open-source licenses. All assets are cited in the text.

Guidelines:

- The answer [N/A] means that the paper does not use existing assets.
- The authors should cite the original paper that produced the code package or dataset.
- The authors should state which version of the asset is used and, if possible, include a URL.

- The name of the license (e.g., CC-BY 4.0) should be included for each asset.
- For scraped data from a particular source (e.g., website), the copyright and terms of service of that source should be provided.
- If assets are released, the license, copyright information, and terms of use in the package should be provided. For popular datasets, `paperswithcode.com/datasets` has curated licenses for some datasets. Their licensing guide can help determine the license of a dataset.
- For existing datasets that are re-packaged, both the original license and the license of the derived asset (if it has changed) should be provided.
- If this information is not available online, the authors are encouraged to reach out to the asset's creators.

13. **New assets**

Question: Are new assets introduced in the paper well documented and is the documentation provided alongside the assets?

Answer: [Yes]

Justification: The new asset is the Bemis–Murcko scaffold-disjoint ChEMBL benchmark (2,699 drug–mechanism pairs with split manifests). Its construction is described in Section 3.1 and 3.3. Code and split manifests will be released upon acceptance.

Guidelines:

- The answer [N/A] means that the paper does not release new assets.
- Researchers should communicate the details of the dataset/code/model as part of their submissions via structured templates. This includes details about training, license, limitations, etc.
- The paper should discuss whether and how consent was obtained from people whose asset is used.
- At submission time, remember to anonymize your assets (if applicable). You can either create an anonymized URL or include an anonymized zip file.

14. **Crowdsourcing and research with human subjects**

Question: Does the paper include the full text of instructions given to participants and screenshots, if applicable, as well as details about compensation (if any)?

Answer: [N/A]

Justification: The paper does not involve crowdsourcing or research with human subjects.

Guidelines:

- The answer [N/A] means that the paper does not involve crowdsourcing nor research with human subjects.
- Including this information in the supplemental material is fine, but if the main contribution of the paper involves human subjects, then as much detail as possible should be included in the main paper.
- According to the NeurIPS Code of Ethics, workers involved in data collection, curation, or other labor should be paid at least the minimum wage in the country of the data collector.

15. **Institutional review board (IRB) approvals or equivalent for research with human subjects**

Question: Does the paper describe potential risks incurred by study participants, whether such risks were disclosed to the subjects, and whether Institutional Review Board (IRB) approvals (or an equivalent approval/review based on the requirements of your country or institution) were obtained?

Answer: [N/A]

Justification: The paper does not involve human subjects. No IRB approval is required.

Guidelines:

- The answer [N/A] means that the paper does not involve crowdsourcing nor research with human subjects.

- Depending on the country in which research is conducted, IRB approval (or equivalent) may be required for any human subjects research. If you obtained IRB approval, you should clearly state this in the paper.
- We recognize that the procedures for this may vary significantly between institutions and locations, and we expect authors to adhere to the NeurIPS Code of Ethics and the guidelines for their institution.
- For initial submissions, do not include any information that would break anonymity (if applicable), such as the institution conducting the review.

16. **Declaration of LLM usage**

Question: Does the paper describe the usage of LLMs if it is an important, original, or non-standard component of the core methods in this research? Note that if the LLM is used only for writing, editing, or formatting purposes and does *not* impact the core methodology, scientific rigor, or originality of the research, declaration is not required.

Answer: [N/A]

Justification: LLMs are not a component of the MoleculeLens method. The text encoder (S-Biomed-RoBERTa) is an encoder-only model used purely for feature extraction without any generative LLM component.

Guidelines:

- The answer [N/A] means that the core method development in this research does not involve LLMs as any important, original, or non-standard components.
- Please refer to our LLM policy in the NeurIPS handbook for what should or should not be described.

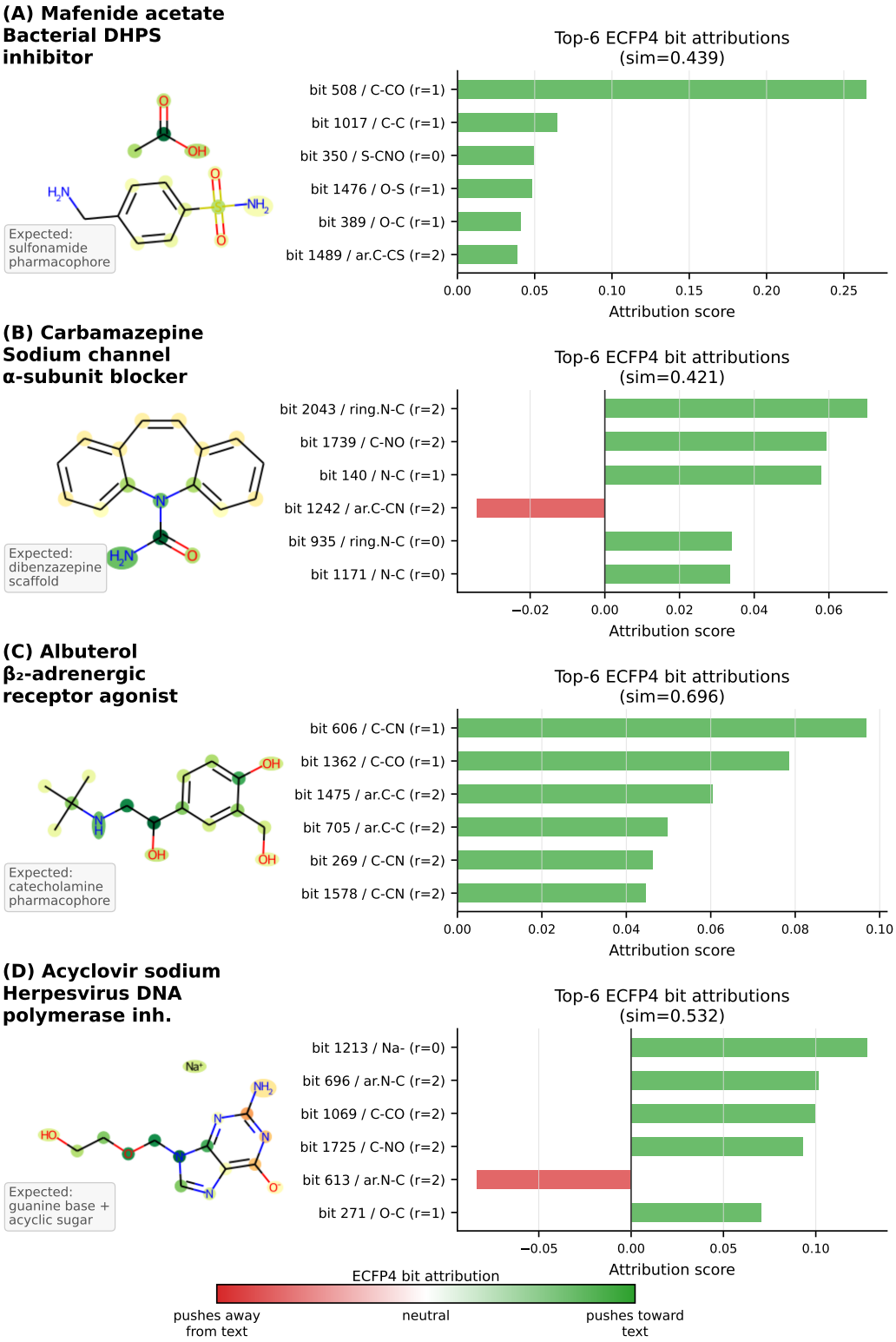

Figure 9: **ECFP4 linear saliency examples for correctly retrieved pairs.** Atoms are colour-coded by ECFP4 attribution value and paired with top-bit bar charts.

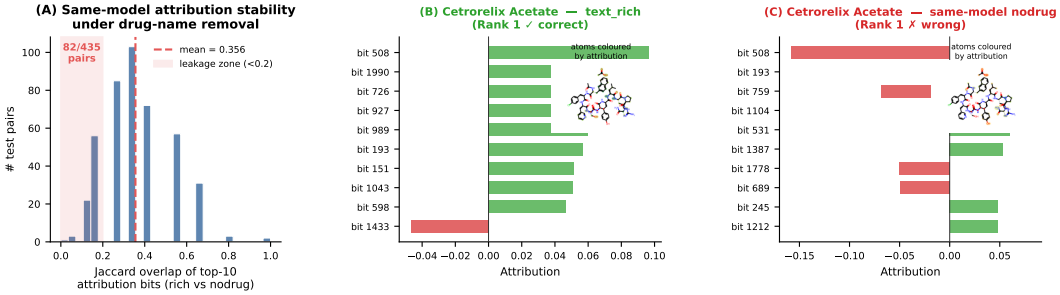

Figure 10: **Same-model drug-name ablation shifts salient substructures.** The distribution of Jaccard overlaps shows that 82 test pairs fall below $J < 0.2$; the Cetrorelix acetate example illustrates how saliency changes when the explicit name is removed.

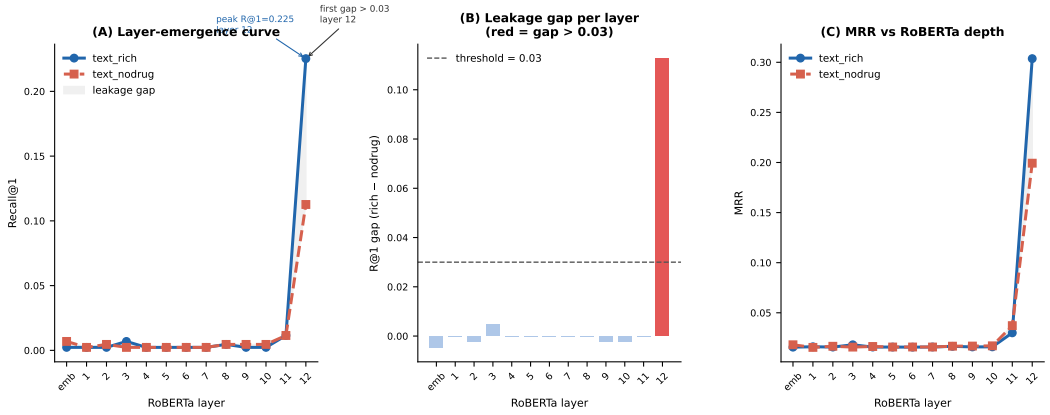

Figure 11: **Contrastive logit lens across RoBERTa layers.** Retrieval is near chance until the final transformer layer, where both rich retrieval signal and the no-name gap emerge.

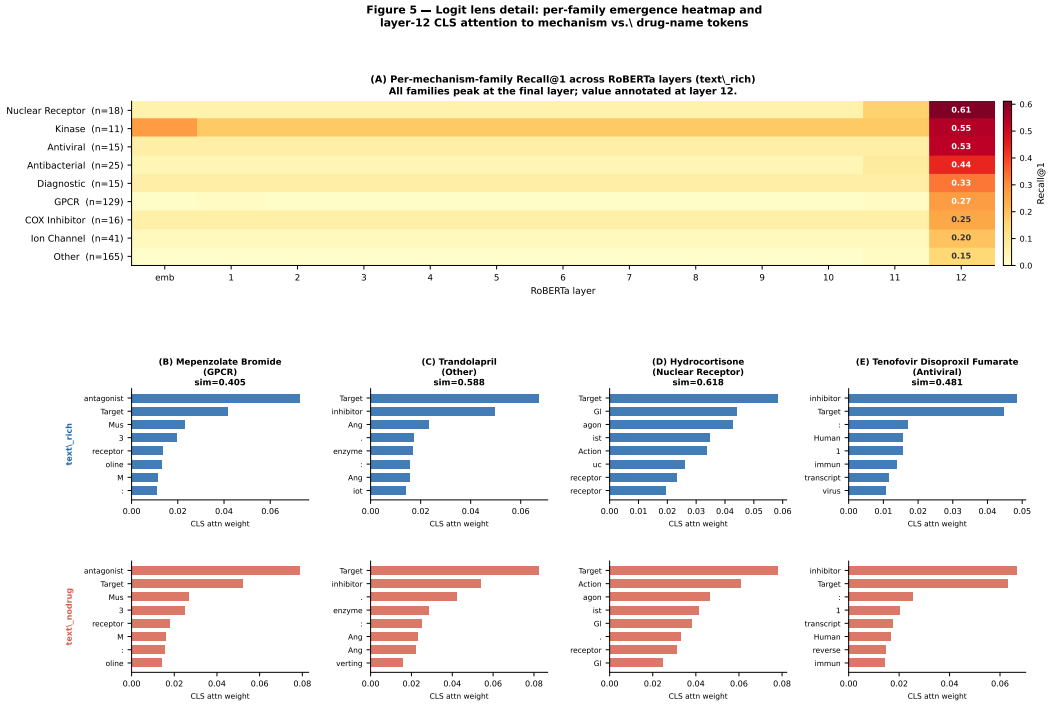

Figure 12: **Logit-lens detail.** Family-level layer-12 retrieval and CLS attention examples under rich and no-drug text conditions.